TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

1. Marcus has \$35 and saves \$15 per week. Which equation gives the amount he has saved, y, after x weeks?

A. $y = 35x + 15$

B. $\boldsymbol{y = 15x + 35}$

C. $y = -15x + 35$

D. $y = 50x$

Start at \$35 (b) and add \$15 each week (m): $y = 15x + 35$. Like test Q1.

2. What value of x makes $3(x - 2) = 12$ true?

F. $4\frac{2}{3}$

G. 2

H. **6**

J. 10

Distribute: $3x - 6 = 12$, so $3x = 18$ and $x = 6$. $4\frac{2}{3}$ forgets to multiply the 2.

3. Is $x = 2$ a solution of $5x - 1 = 9$?

A. **Yes, because 9 = 9**

B. No, because $10 \neq 9$

C. Yes, because $x = 9$

D. No, because $11 \neq 9$

Substitute: $5(2) - 1 = 10 - 1 = 9$. A true statement means 2 is a solution.

4. Which expression is equivalent to $4(x + 2) - x$?

F. $3x + 2$

G. $5x + 8$

H. $4x + 6$

J. $\boldsymbol{3x + 8}$

Distribute: $4x + 8 - x = 3x + 8$.